Extending the Spatio-Temporal Skew-t Process for Threshold Exceedance Prediction

AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

Yi-Xuan Xie

Advisor: Prof. Hao-Yun Huang

Department of Applied Mathematics
National Dong Hwa University

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Outline

Problem background

- Monitoring networks: limited coverage $\rightarrow$ statistical extrapolation, with quantified confidence
- Monitoring and compliance
- Target: $\Pr(Y > L)$ at unmonitored sites — the upper tail, not the mean field

Ozone NAAQS (ppb)	1997	2008	2015 (current)
Annual 4th-highest daily max 8-hr avg., 3-year mean	80	75	70

Why standard spatial methods fall short

1. **Margin** — Gaussian: light-tailed, symmetric; ozone maxima: heavy-tailed, right-skewed
2. **Extremal Dependence** (Coles, Heffernan, and Tawn, 1999):

$$\chi(h) = \lim_{c \to c^*} \Pr[\, Y(\mathbf{s} + h) > c \mid Y(\mathbf{s}) > c\,], \qquad \text{Gaussian: } \chi(h) = 0 \ \ \forall h$$

heavy tails + asymmetry + asymptotic dependence $\Rightarrow$ skew-t process

Why not max-stable or EVT threshold models

- Max-stable likelihood: infeasible at $1{,}089$ stations
- 75 ppb $\approx$ 0.92 sample quantile — not the far tail in EVT sense
- Subasymptotic setting (Huser and Wadsworth, 2022): joint tail at high but *finite* levels
- Skew-t $=$ location-scale mixture of GPs $\in$ this family

Baseline STP: model and parameters

Morris et al. (2017):

$$Y_t(\mathbf{s}) \;=\; \underbrace{\mathbf{X}_t(\mathbf{s})^\top\beta}_{\text{regression mean}} + \underbrace{\lambda\,\sigma_t(\mathbf{s})\,|z_t(\mathbf{s})|}_{\text{skewness term}} + \underbrace{\sigma_t(\mathbf{s})\,v_t(\mathbf{s})}_{\text{spatial process}}$$

- $\lambda \in \mathcal{R}$: skewness $\cdot$ $z_t(\mathbf{s}) \sim N(0,1)$, $|z_t|$ half-normal
- $\sigma_t^2(\mathbf{s}) \sim \mathrm{IG}(a/2,\ b/2)$: scale mixing $\rightarrow$ heavy tails ($a = \text{df}$)
- $v_t(\mathbf{s})$: standard GP, Matérn correlation (φ range, ν smoothness, η spatial proportion)
- Marginalize $|z_t|$, σ_t $\Rightarrow$ skew-t margin + asymptotic dependence

Baseline STP: Morris's extensions

1. **Random partition**, K knots redrawn daily:

$$P_{tk} = \{\mathbf{s} : k = \arg\min_{\ell} \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \qquad z_t(\mathbf{s}) = z_{tk}, \quad \sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$$

   $\rightarrow$ local asymptotic dependence

2. **Censored likelihood**: $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}),\ T\}$, $T \le L$ — upper-tail focus; below T: imputed in MCMC

3. **Temporal dependence**: PIT $\rightarrow$ Gaussian scale $\rightarrow$ stationary AR(1); margins preserved at every t

Two candidate extensions

STP baseline
Morris et al., 2017

Temporal axis: AR(1) $\rightarrow$ standardized AR(2)
longer-range dependence in the record

Spatial axis: MRTS basis in the fixed-effect mean
large-scale variation to the basis, small-scale to Matérn

candidate options, not replacements — held-out scores decide

Extension 1: a standardized AR(2) prior

On the PIT-transformed Gaussian scale:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \qquad (\phi_1, \phi_2) \in \text{stationarity triangle}$$

Inverse PIT needs $\gamma_0 = \mathrm{Var}(Y_t) = 1 \;\Rightarrow\; \sigma_\epsilon^2$ not free (Yule–Walker):

$$\gamma_1 = \frac{\phi_1}{1-\phi_2}, \qquad \gamma_2 = \phi_1\gamma_1 + \phi_2, \qquad \sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2$$

- Initial pair (Y_1, Y_2): stationary bivariate Gaussian, correlation γ_1
- Identical margins across i.i.d. / AR(1) / AR(2)

Extension 2: MRTS basis functions in the mean

MRTS (Tzeng and Huang, 2018):

- Ordered coarse $\rightarrow$ fine: $f_1 = 1$; $f_2, f_3 =$ coordinates; $k \geq 4$: eigenvectors of the TPS kernel, eigenvalues decreasing
- Closed form at *new* locations $\rightarrow$ predictive covariate

$$\mathbf{X}_t^*(\mathbf{s}) = \left[\mathbf{f}(\mathbf{s})^\top, \mathbf{X}_t(\mathbf{s})^\top\right]^\top, \qquad Y_t(\mathbf{s}) = \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* + \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| + \sigma_t(\mathbf{s})\, v_t(\mathbf{s})$$

- Claim: non-stationary mean only — Matérn keeps small-scale dependence

Hierarchical model and inference

Three stages (Berliner, 1996):

Stage 1 censored data: $\widetilde{Y}_t(\mathbf{s}_i) = \max\{Y_t(\mathbf{s}_i), T\}$

Stage 2 Gaussian field · partition · latent AR(2) dynamics

Stage 3 priors; $(\phi_1, \phi_2) \sim$ Unif(stationarity triangle)

$$\left[\mathbf{Y}_{\text{cens}}, \mathcal{Z}, \boldsymbol{\theta} \mid \widetilde{\mathbf{Y}}\right] \propto \prod_{t=1}^{n_t} \left[\widetilde{\mathbf{Y}}_t \mid \mathbf{Y}_t\right] \left[\mathbf{Y}_t \mid \mathcal{Z}_t, \boldsymbol{\theta}\right] \left[\mathcal{Z}_{1:n_t} \mid \boldsymbol{\theta}\right] \left[\boldsymbol{\theta}\right]$$

- Gibbs-within-MH; AR(2) full conditionals: appendix
- K fixed = candidate model index, chosen by held-out Brier

Simulation design

144 sites on $[0, 10]^2$ · 50 days · 10 data sets · 100 train / 44 test

Analysis models	Gaussian; skew-t ($K{=}1, 5$); sym-t, $T{=}q(0.80)$ ($K{=}1, 5$)
AR(2) designs	$(0.80, -0.35)$, $(0.12, -0.05)$ at $K = 1, 5$; persistent $(0.15, 0.80)$; ARFIMA $d = 0.45$
MRTS designs	$2\times$ second-moment non-stationarity; $2\times$ nonlinear mean surface

Evaluation protocols

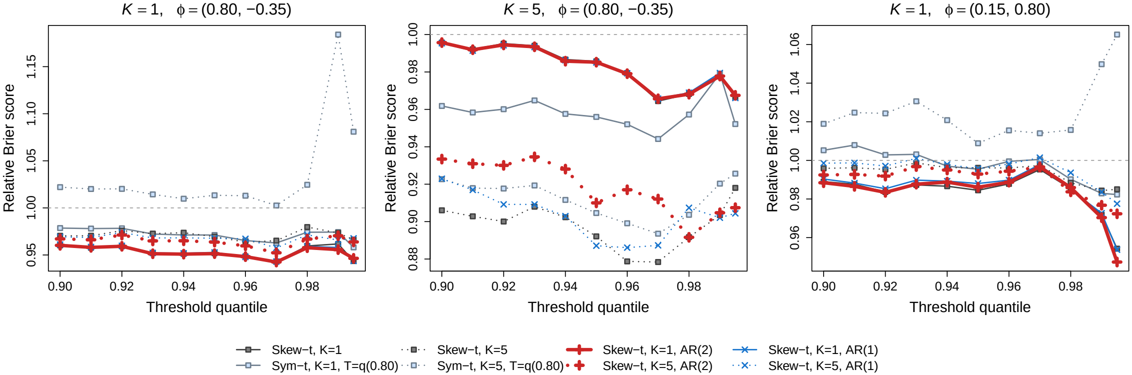
Scoring (Gneiting and Raftery, 2007):

$$\mathrm{BS}(L) = \left\{ I(y > L) - \widehat{P}(Y > L) \right\}^2 \qquad \text{(lower is better)}$$

Protocol	Observed	Predicted	Task
Spatial hold-out	training sites, full record	unseen sites	interpolation
Time-block forecast	all sites, days 1–50	same sites, next 15 days	extrapolation

- Uncertainty: **paired** differences · 95% CI · Wilcoxon

AR(2), interpolation: temporal structure barely moves the scores



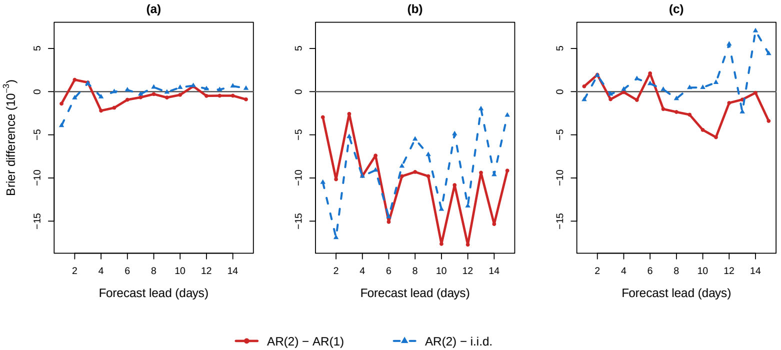
$32/33$ paired intervals cover zero (all six settings: backup)

AR(2): coefficients identified, scores still unmoved

	$\hat{\phi}^{(\sigma)}$		$\hat{\phi}^{(z)}$	
Generating design	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$
Persistent $(0.15,\ 0.80)$	$0.112\,(0.132)$	$\mathbf{0.742}\,(0.117)$	$0.175\,(0.152)$	$\mathbf{0.728}\,(0.112)$
ARFIMA, $d = 0.45$	$0.502\,(0.237)$	$\mathbf{0.200}\,(0.239)$	$0.365\,(0.249)$	$\mathbf{0.156}\,(0.169)$

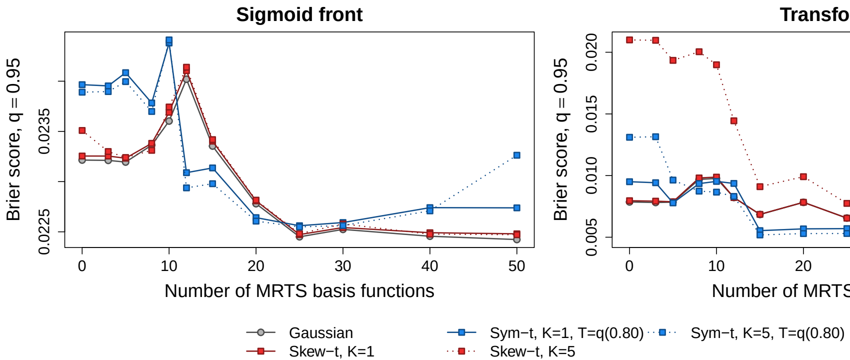
- Second lag recovered — even under long memory
- Hold-out scores: still unmoved $\rightarrow$ the task, not the model

Time-block forecast: the gain tracks persistence



- (a) short memory: nil (b) near-unit-root: -21% at q_{90} ($p = 0.007$) (c) long memory: nil
- Carried by the second lag: AR(1) $\approx$ i.i.d.

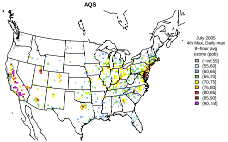
MRTS: inert against second moments, paid off once resolved

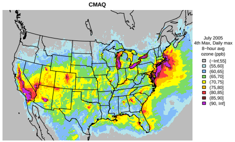


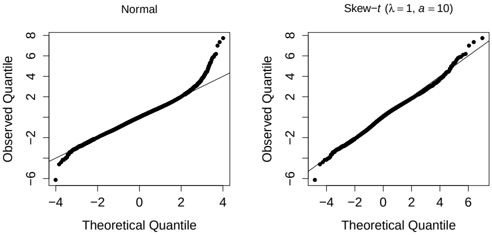
- Second-moment designs: flat — no first moment to recover (backup)
- Nonlinear mean: pays only once the basis resolves the surface; mid-size dip

Ozone data and motivating diagnostics

- July 2005 · AQS daily max 8-h ozone · CMAQ covariate
- Models calibrate CMAQ at the monitors







- Residuals: heavy-tailed, right-skewed
- Skew-t reference $(\lambda = 1,\ a = 10)$: straightens

Candidate set and validation design

- Baseline grid (Morris et al., 2017): {Gaussian, skew-*t* } $\times$ $K \in \{1, \ldots, 15\}$ $\times$ $T \in \{0, 50, 75\}$ ppb $\times$ {TS, no TS}
- Extensions: AR(2) prior; MRTS with $K_{\text{MRTS}} = 10$ pre-specified
- $1{,}089 \to 800$ region-stratified sites $\cdot$ two-fold CV $(400/400)$ $\cdot$ splits $300/100$, $200/200$
- MCMC: $30{,}000$ iterations, $25{,}000$ burn-in

Results: scattered wins, but systematic

Level L	Rank 1 model	Rank 2 model
0.900	**MRTS** skew-t , $K{=}1$, $T{=}0$, AR(1), $K_{\mathrm{MRTS}}{=}10$	skew-t , $K{=}1$, $T{=}0$, AR(1)
0.950	**AR2** skew-t , $K{=}7$, $T{=}50$	**AR2** skew-t , $K{=}10$, $T{=}50$
0.980	skew-t , $K{=}6$, $T{=}50$, AR(1)	skew-t , $K{=}5$, $T{=}50$, AR(1)
0.990	symmetric-t, $K{=}9$, $T{=}75$, AR(1)	skew-t , $K{=}5$, $T{=}50$
0.995	**AR2** symmetric-t, $K{=}15$, $T{=}75$	symmetric-t, $K{=}9$, $T{=}75$, AR(1)

- Deeper tail $\rightarrow$ higher censoring, larger K
- Pattern reproduces on the $300/100$ and $200/200$ splits

Uncertainty: the two axes part ways

Site-clustered bootstrap ($B = 2000$, by fold), $400/400$ split, $\Delta \times 10^3$ [95% CI]:

L quantile	AR(2) − AR(1)	MRTS − none
0.900	-0.15 $[-0.38, +0.08]$	-0.03 $[-0.13, +0.07]$
0.950	-0.12 $[-0.29, +0.04]$	$\mathbf{-0.08}^*$ $[-0.14, -0.02]$
0.980	$+0.03$ $[-0.08, +0.14]$	-0.04 $[-0.07, +0.00]$
0.990	-0.06 $[-0.14, +0.03]$	$\mathbf{-0.03}^*$ $[-0.05, -0.00]$
0.995	-0.03 $[-0.08, +0.02]$	-0.01 $[-0.02, +0.01]$

- AR(2): every interval covers zero (all splits)
- MRTS: negative everywhere; stars survive block resampling; $\approx 3\text{–}4 \times 10^{-3}$ of baseline

Conclusions

- **Temporal axis**: divides along the task — identified yet inert in interpolation; gains only in forecasting, never significant
- **Spatial axis**: real, though modest — needs mean-level structure, and a basis fine enough to resolve it
- **Contribution**: STP baseline $\rightarrow$ candidate model class, selected by held-out threshold prediction

Limitations and future work

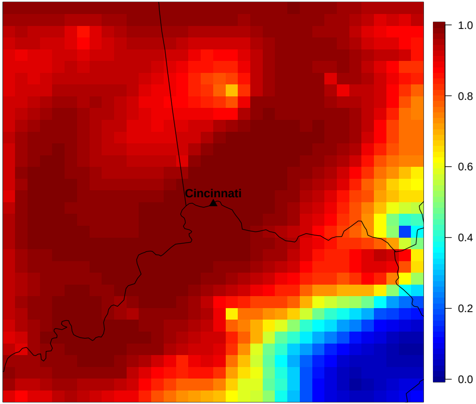
Limitations

- One month, one pollutant · single forecast origin

Future work

- Random-coefficient MRTS (fixed-rank kriging) $\rightarrow$ the second-moment layer
- Reparameterized z , longer records , rolling forecast origin

An exceedance-probability map



$\Pr(\geq 1$ day > 75 ppb), July 2005, Cincinnati — skew-t + TS + MRTS ($K_{\mathrm{MRTS}}=10$)

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Thank You

Questions & Comments Welcome

Backup

Backup: λ–z reflection ridge and forecast screening

- A reflection-type non-identifiability links λ and z: short-record fits can land on the reflected ridge, where the skewness parameter and the intercept fail to separate and the predictive spread inflates.
- The screening rule is public and stated with the forecast protocol: fits whose λ and intercept fail to separate, or whose predictive spread is unbounded, are removed *before* scoring.
- Root fix — a sign-anchored reparameterization of z — is listed as future work.

Backup: AR(2) stationary moments and full conditionals

- Yule–Walker under $\gamma_0 = 1$: $\gamma_h = \phi_1\gamma_{h-1} + \phi_2\gamma_{h-2}$ for $h \geq 1$ gives $\gamma_1 = \phi_1/(1-\phi_2)$, $\gamma_2 = \phi_1\gamma_1 + \phi_2$, and $\sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2$.
- Gibbs-within-MH: the full conditional of Y_t carries three prior factors,

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t)\ \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}}\ \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}}\ \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}};$$

  omitting the lag-2 factor distorts the stationary distribution.
- Head of the series ($t = 1, 2$): the stationary initial-pair density replaces the self factor. Proposals: symmetric Gaussian random walk tuned to 0.3–0.4 acceptance; (ϕ_1, ϕ_2) updated jointly under the uniform prior on the stationarity triangle.

Backup: second-moment design — "inert" is not "no signal"

- The random effect superposes two fixed cosine fields with daily-redrawn Gaussian weights:

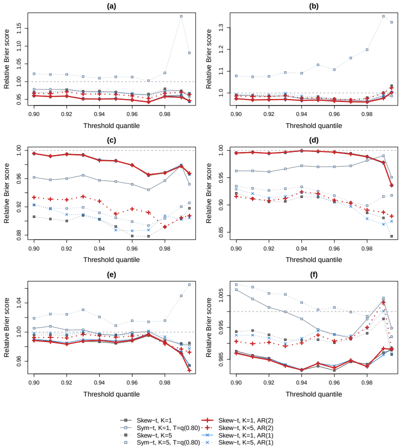
$$u_t(\mathbf{s}) = \sum_{j=1}^{2} \xi_{tj}\, g_j(\mathbf{s}), \quad \xi_{tj} \overset{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3),$$

  giving the location-dependent covariance $\mathrm{Cov}\{u_t(\mathbf{s}), u_t(\mathbf{s}')\} = \sum_j \tau_j^2 g_j(\mathbf{s}) g_j(\mathbf{s}')$.
- Zero time-average $\Rightarrow$ the pooled coefficient has no first-moment structure to recover, while u_t still contributes substantial non-stationary variance: structurally out of reach for a fixed-effect mean, not absent signal.
- The two nonlinear-mean surfaces were confirmed by an oracle calculation before any fitting to leave genuine Brier headroom.

Backup: censoring pairing and MCMC settings

- Why symmetric-t with censoring: once the lower tail is censored, the skewness parameter is only weakly identified, so censored settings pair with a symmetric-t marginal, following Morris et al. (2017).
- Thresholds: the censor level $T = 75$ ppb sits below the era standard $L = 80$ ppb ($T \leq L$), and every evaluation level lies above T; the degrees of freedom a control the tail thickness.
- Burn-in $25{,}000$ of $30{,}000$ ($\approx 83\%$): mixing is slowed by censoring imputation and the latent volume; traces were inspected, and the long burn-in is a conservative choice.

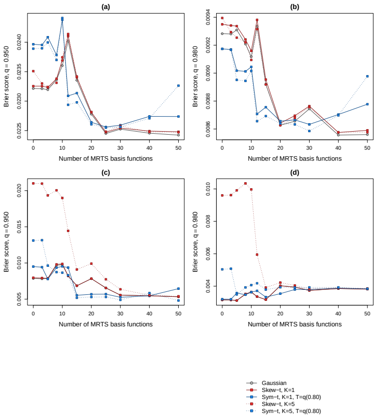
Backup: spatial hold-out, all six generating settings



Relative Brier score vs. the Gaussian reference ($\equiv 1$, dashed).

- (a)–(d): fixed AR(2) designs, $K \in \{1, 5\}$, $(\phi_1, \phi_2) = (0.80, -0.35)$ or $(0.12, -0.05)$.
- (e): persistent $(0.15, 0.80)$.
- (f): ARFIMA, $d = 0.45$ (only the K=1 AR(1)/AR(2) fits).

Backup: MRTS nonlinear-mean designs, full grid



Brier score vs. K_{MRTS} ($K_{\mathrm{MRTS}} = 0$ is the no-MRTS baseline).

- (a, b): sigmoid front, $q = 0.95$ / 0.98.
- (c, d): transform mix, $q = 0.95$ / 0.98.
- At $q = 0.98$ the gain is carried by the five-knot models — the sharpness–extrapolation signature.